INCH-POUND

MIL-DTL-32628
11 April 2019

DETAIL SPECIFICATION

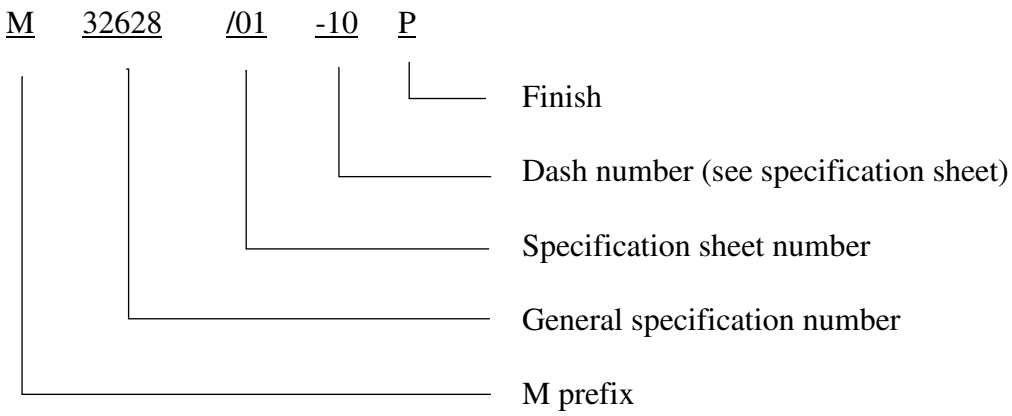
BACKSHELL, ELECTRICAL CONNECTOR, SPRING BAND, ACCESSORY, GENERAL SPECIFICATION FOR

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design and test requirements for an adjustable, backshell spring band, intended for use on electrical connector backshells or shield splice. The backshell spring band provides 360-degree constant pressure around the connector backshell to attach braided shielding to the connector backshell or shield splice.

1.2 Part or Identifying Number (PIN). The PINs to be used for the connector backshell spring band acquired in accordance to this specification are created as follows:



Comments, suggestions, or questions on this document should be addressed to Commander, Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTION

A-A-59569	-	Braid, Wire (Copper, Tin-Coated, Silver-Coated, or Nickel Coated, Tubular or Flat)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32628/1	-	Backshell, Electrical Connector, Tri-Dent Spring Band
MIL-DTL-32628/2	-	Backshell, Electrical Connector, Wide Spring Band
MIL-DTL-32628/3	-	Backshell, Electrical Connector, Wide Spring Band, Heavy Duty
MIL-DTL-32628/4	-	Backshell, Electrical Connector, Wide Spring Band, Tri-Dent Heavy Duty

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A666	-	Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar

(Copies of this document are available online at http://www.astm.org.)

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA-364-54	-	Magnetic Permeability Test Procedure For Electrical Connectors, Contacts, and Sockets

(Copies of this document are available online at http://ecianow.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

SAE INTERNATIONAL

SAE AMS5876 - Cobalt Alloy, Corrosion and Heat-Resistant, Strip, 20Cr-15Ni-40Co-7.0Mo-16Fe, Vacuum Induction Plus Consumable Electrode Remelted, Solution Heat Treated and Cold Rolled

(Copies of this document are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), spring band backshell samples shall be subjected to first article testing in accordance with 4.3.

3.3 Spring band. The coiled spring band shall consist of a stainless steel band in accordance with ASTM A666 or non-magnetic cobalt alloy in accordance with SAE AMS5876 (see table I). The spring band as received shall be coiled with a minimum of three wraps of the spring band material.

3.4 Materials. The materials used in the manufacture of the spring band shall be such that the spring band can meet all the performance and test requirements of this specification and the applicable specification sheet. The spring band shall be so constructed and finished that it shall be free from all defects that may affect proper functioning in service

3.5 Design and construction. The design and construction of the spring band shall be as specified herein. Design shall conform to all performance requirements in 3.7. The design of the coiled spring band shall be specified in the specification sheet.

3.5.1 Spring band design. Spring band shall be designed to accommodate installation on compatible backshells and shield split support ring to secure cable shielding. The coiled spring band shall be capable of being un-coiled and wrapped around the backshell to secure shielding to the connector backshell. The spring band shall secure the shielding 360 degrees around the connector backshell or shield splice and provide sufficient force to hold the shielding in place and provide electrical connectivity between the shielding and the connector backshell or shield splice. The spring band shall be capable of re-installation for a minimum of 25 times and then perform the braid retention test (see 4.6.2.a).

3.6 Dimensions and tolerances. Dimensions for the various sizes of spring bands are as specified in the applicable specification sheet.

3.7 Performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

3.7.1 Magnetic permeability. When tested in accordance with 4.6.1, the relative permeability of the spring band shall be less than 5.0 Mu for non-magnetic cobalt alloy. This requirement is not applicable to ASTM A666 stainless steel spring bands (see applicable specification sheet).

3.7.2 Spring band termination. When tested in accordance with 4.6.2, the spring band slippage shall not exceed .025 inch and the electrical resistance shall not exceed 1.0 milliohm.

3.7.3 Marking. Markings shall include the PIN and the manufacturing company Commercial and Government Entity (CAGE) code and shall be marked on the spring band. Markings shall be clearly readable after all specified performance testing of this specification. Markings shall be examined in accordance with 4.6.3.

3.7.4 Workmanship. Workmanship shall ensure that the spring bands are of uniform quality, and free of defects that would affect their service.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at temperatures of 59 °F to 95 °F (15 °C to 35 °C) and at ambient air pressure and humidity. Whenever these conditions must be closely controlled in order to obtain reproducible results, for referee purposes, a temperature of 77 °F, +0 °F, -3.6 °F (25 °C, +0 °C, -2 °C), relative humidity of 50±2 percent, and atmospheric pressure of 650 to 800 millimeters of mercury shall be used.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.3 First article inspection. First article inspection shall be performed on backshell/shield braid and spring band assemblies. This inspection shall include examinations and tests specified in table I.

4.3.1 First article inspection samples. See applicable specification sheet for required sample assemblies, and spare backshells, shield braids and spring bands. The shield braid shall be a six-inch braid supplied with each assembly. The braid shall be tin coated copper, round shape, in accordance with A-A-59569 with 90 percent coverage unless otherwise specified in the applicable specification sheet.

TABLE I. First article inspections and tests.

Examinations and Tests	Requirement	Test Methods
Visual and dimensional examination	3.3, 3.4, 3.5, 3.7.4	4.5.1
Magnetic permeability 1/	3.7.1	4.6.1
Spring band termination	3.7.2	4.6.2
Product marking and legibility	3.7.3	4.6.3

1/ Test not required for ASTM A666 stainless steel spring bands.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection. The conformance inspections shall include the following inspections:

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of conformance inspection tests specified in table II.

4.4.1.1 Inspection lot. An inspection lot shall consist of spring band for a particular size produced under essentially the same manufacturing conditions and presented for inspection at one time.

4.4.1.2 Sampling plan. Statistical sampling shall be in accordance with table III. A randomly selected group(s) of samples is required. For acceptance of the lot there shall be zero occurrences of defects.

TABLE II. Conformance inspections.

Examination and tests	Requirement paragraph	Test paragraph	Sampling plan
Visual and dimensional examination	3.3, 3.4, 3.5, 3.7.4	4.5.1	4.4.1.2
Magnetic permeability 1/	3.7.1	4.6.1	
Product marking and legibility	3.7.3	4.6.3	

1/ Test not required for ASTM A666 stainless steel spring bands.

TABLE III. Zero defect sampling plan.

Lot size	Minimum number of spring bands to be tested
1 – 4	All
5 – 50	5
51 – 90	7
91 – 150	11
151 – 280	13
281 – 500	16
501 – 1200	19
1201 – 3200	23
3201 – 10000	29
10001 – 35000	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

4.5 Examinations.

4.5.1 Visual and dimensional examination. The spring band shall be given a thorough mechanical and visual examination to determine that all materials and workmanship comply with the specified requirements. Dimensions shall be measured with calibrated devices using standard laboratory methodology. Material shall be verified by manufacturer certificate of materials.

4.6 Test methods.

4.6.1 Magnetic permeability (3.7.1). The relative permeability shall be tested in accordance with ECIA EIA-364-54. The spring band may be tested on a non-magnetic mandrel or as part of a connector back shell assembly.

4.6.2 Spring band termination (3.7.2). Perform the following tests on four complete assemblies for each backshell shell size for a total of twelve specimens (see 4.3.1). The six-inch braid shall be installed using the spring band to the backshell. The spring band shall be installed on the backshell using manufacturer's recommended installation procedures. The shield braid shall not extend more than .125 inch from the connector side of the spring band.

a. Braid retention. One complete assembly consisting of backshell, shield braid and spring band for each size shall be tested. With the backshell suitably assembled with braid and spring band, pull the braid at a rate of one inch per minute to a force of 100 pounds for braid .50 inch inner diameter and under, and 150 pounds for braid over .50 inch inner diameter unless otherwise specified in the applicable specification sheet. The braid shall not pull out. Spring band slippage shall not exceed .025 inch when measured from a fixed point. Braid breakage during testing shall not be considered as a failure.

b. Thermal aging. One complete assembly consisting of backshell, shield braid and spring band for each size shall be tested. The backshell assembly shall be heated to 302 °F (150 ºC) for 168 hours in a thermal chamber followed by an electrical resistance measurement at room temperature. The applied current shall be 100 milliamps ±10 milliamps at a maximum of 1.50 DC volts. The measurement shall be taken from a point on the braid, within 1.0+0.00/-0.50 inch beyond the end of the backshell, and a point on the backshell at the opposite side of the band. The electrical resistance shall not exceed 1.0 milliohm.

c. Re-installation testing. One complete set of assemblies for each backshell shield braid and spring band for each size shall be tested for braid retention and another complete set of assemblies for each backshell size shall be tested for thermal aging after installation and re-installation of the spring band in accordance with re-installation times of 3.5.1. A new six-inch braid shall be used for the final installation before testing.

4.6.3 Product marking and legibility (see 3.7.3). Product identification markings shall be visually examined for legibility. Visual examination shall be conducted by uncoiling spring band as needed to view markings. Magnification is not required, but caution should be taken not to bend spring band backward which may result in damage to the spring band.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The connector backshell spring bands are intended for use in terminating individual or overall EMI/RFI shielding to connector backshells. The spring bands are designed to be used in high vibration, shock, highly corrosive, and high temperature environments unique to military applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

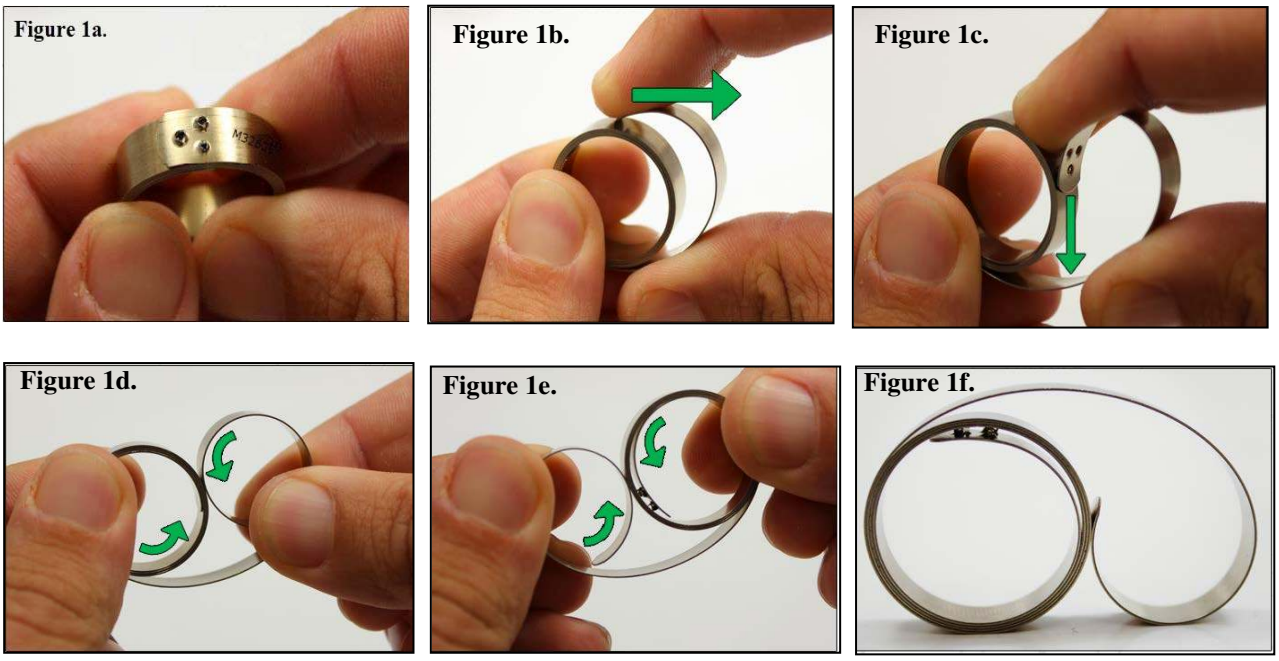
a. Title, number, and date of the specification.
b. Connector backshell spring band part number (see 1.2).
c. When first article inspection is required (see 3.2).
d. Packaging (see 5.1).
e. Minimum package identification information should include: Nomenclature, part identifying number, CAGE code, date manufactured, lot manufactured, and quantity.

6.3 Installation guidance.

6.3.1 Tri-Dent spring band installation. See MIL-DTL-32628/1 and /4.

6.3.1.a Spring preparation. The spring band is installed/removed using no tools. It is installed/removed by using fingers to unravel the band. As the spring band is almost unraveled, there is a potential of springing off the banding platform of the backshell. Support and secure the spring band so that it does not spring off. For installation, prepare the spring band by uncoiling it into two coils (see figures 1a through f). As shown on figure 1b, the spring band should be uncoiled with the Tri-Dents internal to the coil as shown on figures 1c. through 1f. Caution should be taken not to bend spring band backward, which may result in damage to the spring band.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.



FIGURES 1a. - f. Spring installation.

6.3.1.b Shield installation. Prior to installing shield on backshell, it is recommended to clean the backshell termination area with isopropyl alcohol in accordance with TT-I-735 (or equivalent). After installation of the shield braid on the backshell, the shield braid should be compressed uniformly so that it conforms tightly to the profile of the backshell shield termination area (see figure 2) using an AT508K, or equivalent connector plier (see figure 3).



FIGURE 2. Compressing shield braid.



FIGURE 3. Connector pliers (Cannon part number AT508K or equivalent).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

6.3.1.c Spring installation. Press the spring band (see figure 4) against the desired installation location, over the metallic braid and the backshell termination area. Completely uncoil the spring band around the backshell and ensure that it is properly positioned, ensuring the Tri-Dent raised protrusions are on the outer /exposed surface of the spring band (see figure 5). Re-position as required. Trim shield braid to about .125 inch of the connector side of the spring (see figure 6).



FIGURE 4. Spring installation.



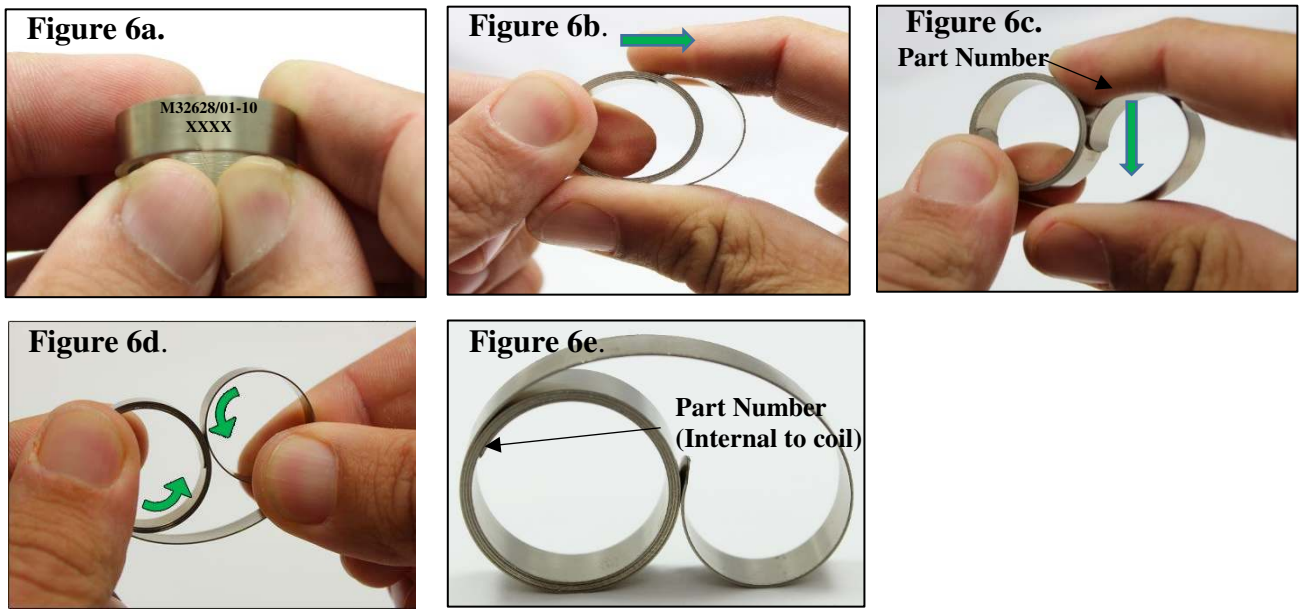
FIGURE 5. Final spring location.



FIGURE 6. Spring/shield braid trimmed.

6.3.2 Wide and Heavy-Duty Wide spring band installation. See MIL-DTL-32628/2 and /3.

6.3.2.a Spring preparation. The spring band is installed/removed using no tools. It is installed/removed by using fingers to unravel the band. As the spring band is almost unraveled there is a potential of springing off the banding platform of the backshell. Support and secure the spring band so that it does not spring off. For installation, prepare the spring band by uncoiling it into two coils (see figures 6a through 6e). As shown in figure 6b, the spring band should be uncoiled with the part number internal to the coil as shown in figures 6c. through 6e. Caution should be taken not to bend spring band backward which may result in damage to the spring band.



FIGURES 6a - e. Spring installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

6.3.2.b Shield installation. Prior to installing shield on backshell, it is recommended to clean the backshell termination area with isopropyl alcohol in accordance with TT-I-735 (or equivalent). After installation of the shield braid on the backshell, the shield braid should be compressed uniformly so that it conforms tightly to the profile of the backshell shield termination area (see figure 7) using an AT508K, or equivalent connector plier (see figure 8).



FIGURE 7. Compressing shield braid.



FIGURE 8. Connector pliers (Cannon part number AT508K or equivalent).

6.3.2.c Spring installation. Press the spring band (see figure 9) against the desired installation location, over the metallic braid and the backshell termination area. Completely uncoil the spring band around the backshell and ensure that it is properly positioned, ensuring the part number is on the outer /exposed surface of the spring band (see figure 10). Re-position as required. Trim shield braid to about .125 inch of the connector side of the spring (see figure 10).



FIGURE 9. Spring installation.



FIGURE 10. Spring/shield braid trimmed.

6.4 Backshell compatibility. Spring band part number compatible with backshells and shield splice are specified in the applicable specification sheet. Using the spring band on other backshells or accessories may not provide the specified performance requirements of MIL-DTL-32628 spring bands. The backshell or accessory shield termination area configuration (finish, knurling, banding shoulder geometry and dimension, etc.) in addition to the OD are important factors in proper shield termination.

6.5 Sensitive circuits applications. The use of the Non-magnetic cobalt alloy spring band is recommended in sensitive circuit applications (see SAE AS50881) to minimize any potential impacts due to any stray magnetic forces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

6.6 First article test waiver consideration. The acquiring activity may choose to waive the thermal aging (4.6.2.b) and re-installation (4.6.2.c) test portions of the first article inspection (Table I) if the manufacturer has completed first article testing within 24 months and provides certification that there has been no change in material and manufacturing processes, and meets the applicable specification sheet requirements.

6.7 Re-installation. Re-installation of the spring band has been validated for a minimum of 25 re-installations. Ensure the spring band meets the following before re-installation.

a. Ensure the correct spring band (part number M32628/XX-XX) is installed on connector accessory (see appropriate specification sheet).

b. Ensure part number is visible. This excludes the smallest size spring band (see appropriate specification sheet), which could show only a partial part number.

c. Ensure “tri-dent” feature, if applicable, is visible.

d. Ensure a close, tight, secure fit around the circumference of the EMI shield as installed on the banding platform of the connector accessory (see 6.3.1.b and 6.3.2.b).

e. If any items (a through d above) are not met, replace with new spring band.

6.8 Subject term (key word) listing.

Conductivity
Magnetic permeability
Shield braid
Shield split support ring

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy-AS

Project 5935-2019-014

Review activities:
Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:
SAE AE-8C1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.